

Deconstructing HCI & CG in Daily Software

Google Earth: Burj Khalifa, Dubai

SLIDE 1: HCI FOCUS

HCI focuses on how users interact with software through interface elements, controls, and feedback mechanisms.

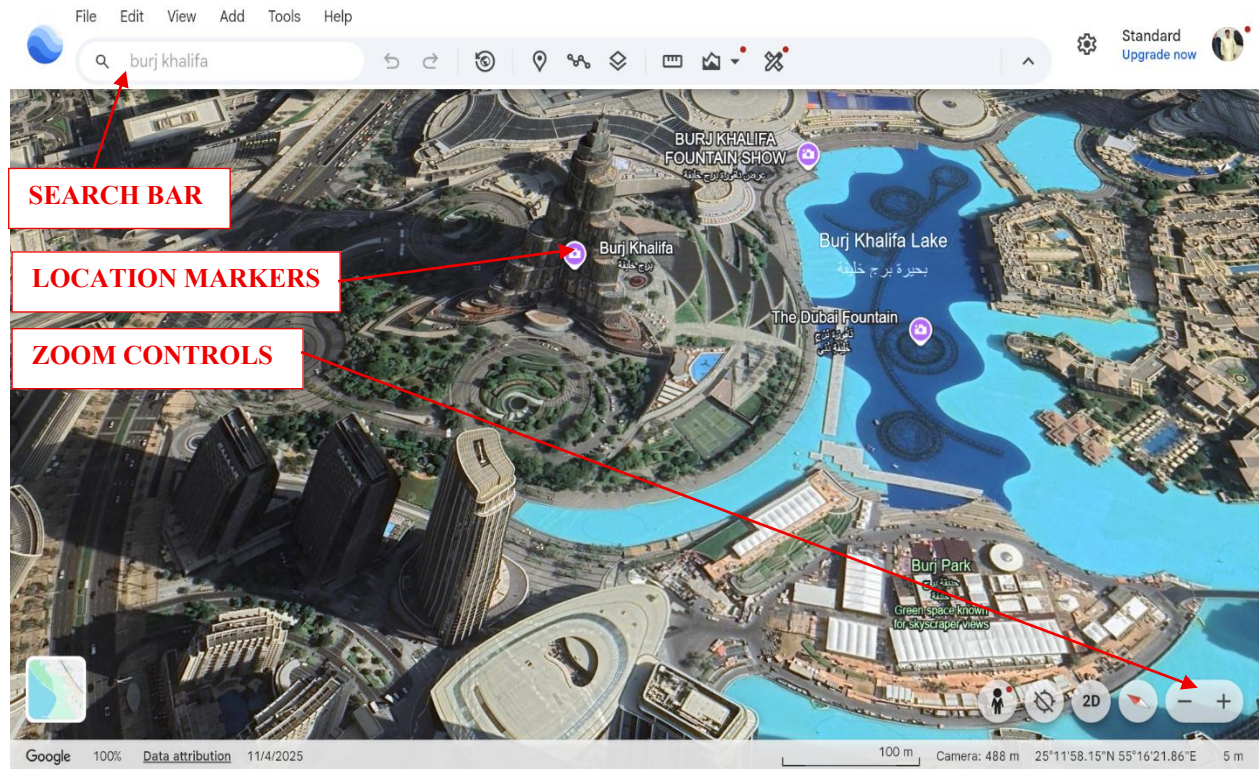


Figure 1: HCI Elements - Interactive Components

1. SEARCH BAR

A text input field in the top-left corner that enables location discovery. When users type location names, the system provides autocomplete suggestions and instantly navigates to the searched location, reducing typing effort and providing clear visual feedback.

2. LOCATION MARKERS

Interactive clickable pins that indicate places of interest. These markers demonstrate excellent affordance design—their appearance suggests they are clickable. Hovering or clicking displays relevant information, creating a conversational interface for visual exploration.

3. ZOOM & ROTATE CONTROLS

Located on the right-side toolbar with +/- buttons for zoom and arrow buttons for rotation. These controls support multiple interaction methods (buttons and mouse gestures), accommodating different user preferences and enabling both first-time exploration and power-user efficiency.

SLIDE 2: COMPUTER GRAPHICS FOCUS

Computer Graphics encompasses visual rendering techniques that create the 3D illusion. These include texture mapping, lighting, perspective projection, and depth cues.

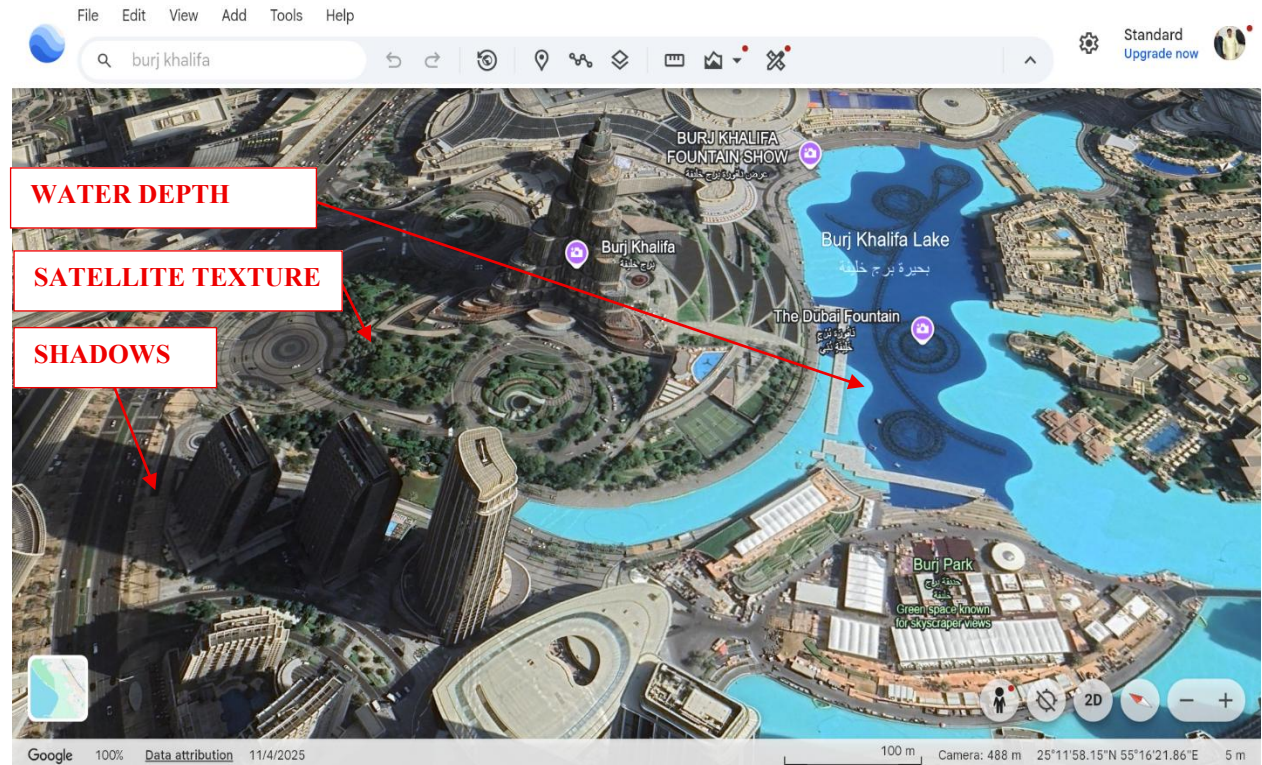


Figure 2: CG Elements - Visual Rendering Components

1. SATELLITE TEXTURE MAPPING

Applies high-resolution satellite photographs to 3D geometric models of Earth's surface. This creates photorealism without modeling billions of individual details. Multiple detail levels (LOD) load based on zoom level, enabling seamless navigation from planetary to street-level views while maintaining performance. Users recognize actual landmarks, improving orientation and usability.

2. SHADOW RENDERING

Calculates shadows cast by buildings based on simulated light sources. Shadows are critical for depth perception—they communicate building height and elevation. Without shadows, the scene appears flat. Shadow length also indicates time of day, adding temporal realism to the visualization.

3. DEPTH CUES & PERSPECTIVE

Uses orthographic projection (parallel projection) for overhead map views, which maintains constant object sizes and simplifies mental maps. Water depth variations, elevation shading, and color gradients create spatial awareness. This projection choice supports HCI goals by matching user mental models of geographic space.

REFLECTION QUESTIONS

Q1: How does Computer Graphics quality help or hinder user interaction?

Graphics quality significantly enhances usability. Photorealistic satellite imagery provides immediate spatial context, helping users recognize landmarks intuitively. The 3D rendering and shadow rendering communicate building heights and spatial relationships that would be ambiguous in 2D maps. The orthographic camera perspective combined with depth cues helps users understand scale and distance relationships.

Minor hindrance: High-resolution textures can cause performance lag on slower devices during rapid panning or zooming, briefly displaying low-resolution imagery. However, the usability benefits far outweigh this minor drawback.

Q2: What is one simple HCI improvement you would make?

Proposed Improvement: Add a "Reset View" button to the toolbar.

Currently, when users explore various locations and rotate the view through multiple interactions, they must manually re-orient themselves to return to their original view. A home icon (△) or "Reset" button on the right toolbar would instantly return the view to a default state: centered on the user's home location, at standard zoom, with north-pointing orientation.

HCI Benefits: (1) Reduces user action count and cognitive effort; (2) Visible button ensures discoverability without menu diving; (3) Enables error recovery—users can quickly undo disorientation; (4) Improves user confidence, especially for less-experienced users.

CONCLUSION

Excellent software design requires careful attention to both HCI and Computer Graphics. Google Earth succeeds because realistic graphics provide spatial context that makes interaction intuitive, while well-designed controls enable users to fully appreciate the visual capabilities. These two aspects work synergistically to create an effective and engaging user experience.

NAME : ASHFAQUE AHMED
ROLL NO : 2K24/CSE/38